UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (10 MARKS)

Answer ALL questions.

1. What does endl do?
   A. Declares an integer
   B. Writes a newline and flushes the stream
   C. Ends the program
   D. Reads a line

2. What is displayed by the following code?

```
int a[4] = {3, 6, 9, 12};
cout << a[1] + a[3];
```

   A. 12
   B. 9
   C. 18
   D. 15

3. Which statement immediately exits a loop?
   A. return 0
   B. cin
   C. continue
   D. break

4. Given the declaration, what is displayed?

```
char word[6] = {'p', 'r', 'i', 'n', 't', '\0'};
cout << word[3];
```

   A. i
   B. p
   C. t
   D. n

5. Which of the following are valid identifiers?
   i. total_score
   ii. 2total
   iii. _count
   iv. int
   A. iii and iv
   B. ii and iv
   C. i and ii
   D. i and iii

6. Which term describes a finite sequence of precise steps that solves a problem?
   A. Compiler
   B. Variable
   C. Array
   D. Algorithm

7. Which values can a bool variable hold?
   A. any integer
   B. true or false

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C. any character
D. any decimal

8. How many times does a while loop execute when its condition is false initially?
A. Until the user stops it
B. Zero
C. Two
D. One

9. Which of the following are selection structures?
i. if
ii. switch
iii. for
iv. while
A. ii and iv
B. iii and iv
C. i and iii
D. i and ii

10. Which item is NOT a programming paradigm?
A. Debugging
B. Structured programming
C. Object-oriented programming
D. Procedural programming

PART B (60 MARKS)

QUESTION 1

This question tests algorithms and programming errors.

a) Define algorithm and pseudocode.

(3 marks)

b) Explain THREE types of programming errors and give one example of each.

(2 marks)

QUESTION 2

A program reads a student's score from 0 to 100 and displays the grade shown in Table 1.

Table 1. Grade Scale

Score	Grade
80-100	A
60-79	B
50-59	C
40-49	D
0-39	F

a) Draw a flowchart for the grade classification.

(8 marks)

b) Write pseudocode for the algorithm and convert it into a complete C++ program.

(5 marks)

QUESTION 3

This question tests operator precedence and assignment.

a) Evaluate the following expressions using the declarations in the code block.

```
int a = 12, b = 5;
double c = 2.5;
```

(5 marks)

b) Write C++ statements to declare two integers a and b, read their values, and display their sum, difference, product and remainder.

(6 marks)

QUESTION 4

Study the following C++ program segment:

```
int total = 0;
for (int i = 1; i <= 5; i++) {
    total += i;
}
cout << total;
```

a) Trace the program and state the output.

(10 marks)

b) Rewrite the loop using a while loop. Then explain what changes if the condition is i < 5.

(5 marks)

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QUESTION 5

A sports club records the sport names and medal counts in Table 1.

a) Declare two parallel arrays for the five sports in Table 1 and initialise them with those values.

Table 2. Sport and Medal Records

Sport	Medals
Swimming	8
Badminton	12
Athletics	7
Cycling	5
Hockey	9

(6 marks)

b) Write a program segment to display each sport with its medals, then calculate the total and average medals.

(10 marks)

PART C (30 MARKS)

QUESTION 1

A lecturer records the marks of ten students in an array.

Write a complete C++ program that reads ten marks, displays the average, the highest mark and its index, and counts how many marks are above the average.

(15 marks)

QUESTION 2

A catering company charges the meal prices shown in Table 1.

Table 3. Meal Price

Quantity Range	Price per Meal (RM)
50 - 99	11.50
100 - 199	10.50
200 and above	9.50

Write a complete C++ program that reads the customer name and quantity, validates the quantity, calculates the total, displays a receipt, and repeats until the user chooses to stop. At the end, display the total quantity and total sales.

(15 marks)

END OF QUESTION PAPER

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